

# J.J. EMERSON

UNIVERSITY OF CALIFORNIA IRVINE

DEPARTMENT OF SYSTEMS BIOLOGY

DEPARTMENT OF ECOLOGY AND EVOLUTIONARY BIOLOGY

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## EDUCATION/ACADEMIC POSITIONS

|           |                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Current   | Professor, Dept. Systems Biology, Dept. Ecology & Evolutionary Biology, University of California Irvine                           |
| 2020-2025 | Associate Professor, Dept. Ecology & Evolutionary Biology, Center for Complex Biological Systems, University of California Irvine |
| 2013-2020 | Assistant Professor, Dept. Ecology & Evolutionary Biology, Center for Complex Biological Systems, University of California Irvine |
| 2010-2013 | Postdoctoral Fellow, Integrative Biology, University of California Berkeley (Advisor: Doris Bachtrog)                             |
| 2006-2010 | Postdoctoral Fellow, Genomics Research Center, Academia Sinica, Taipei, Taiwan (Advisor: Wen-Hsiung Li)                           |
| 2000-2006 | PhD Ecology & Evolution, University of Chicago (Advisor: Manyuan Long)                                                            |
| 1996-2000 | BA Biochemistry/Ecology & Evolution, Rice University                                                                              |

PREPRINTS: <http://emersonlab.org/publications/#preprints>

## PEER REVIEWED PUBLICATIONS (REVERSE CHRONOLOGICAL ORDER, \* INDICATES EQUAL CONTRIBUTION,

✉ INDICATES CORRESPONDING AUTHOR)

41. Daigle A., Whitehouse L.S., Zhao R., **Emerson J.J.**, Schrider D.R. ✉. Leveraging long-read assemblies and machine learning to enhance short-read transposable element detection and genotyping. *Genetics* (2026) iyag101.
40. Hu M.\*, P Wan P.\*, Chen C.\*, Tang S, Chen J, Wang L, Chakraborty M, Zhou Y., Chen J., Gaut B.S., **Emerson J.J.** ✉, Liao Y. ✉. Accurate, scalable structural variant genotyping in complex genomes at population scales. *Molecular Biology and Evolution* (2025) msaf180.
39. Shukla H.G., Chakraborty M. ✉, **Emerson J.J.** ✉. Genetic variation in recalcitrant repetitive regions of the *Drosophila melanogaster* genome. *Genome Research* (2025) 35: 2023-2040.
38. Xia S., Chen J., Arsala D., **Emerson J.J.**, Long M. Functional innovation through new genes as a general evolutionary process. *Nature Genetics* (2025) 1-15.
37. Lee U.J., Arsala D., Xia S., 10 others, **Emerson J.J.**, Andolfatto P., Zhou Q., Zhao L., Long M. The three-dimensional genome drives the evolution of asymmetric gene duplicates via enhancer capture-divergence. *Science Advances* (2025) 10 (51), eadn6625.
36. Daigle A., Whitehouse L.S., Zhao R., **Emerson J.J.**, Schrider D.R. Leveraging long-read assemblies and machine learning to enhance short-read transposable element detection and genotyping. *bioRxiv* (2025) 2025.02.11.637720.
35. Shukla H.G., Chakraborty M. ✉, **Emerson J.J.** ✉. Genetic variation in recalcitrant repetitive regions of the *Drosophila melanogaster* genome. *bioRxiv* (2024) 2024.06.11.598575. [In Review: **Genome Research**]
34. Chakraborty M., Lara A.G., Dang A., McCulloch K.J., Rainbow D., Carter D., Ngo L.T., Solares E.A., Said I., Corbett-Detig R., Gilbert L.E., **Emerson J.J.** ✉, Briscoe A. ✉. Sex-linked gene traffic underlies the

- acquisition of sexually dimorphic UV color vision in *Heliconius* butterflies. **Proc. Natl. Acad. Sci. U.S.A.** (2023) 120, e2301411120.
33. Zhao R. , Lukacsovich T., Gaut R., **Emerson J.J.**  FREQ-Seq<sup>2</sup>: a method for precise high-throughput combinatorial quantification of allele frequencies. **G3: Genes, Genomes, Genetics** (2023) jkad162.
  32. Liao Y.\*, Wang J.\*, Zhu Z., Liu Y., Chen J., Zhou Y., Liu F., Lei J., Gaut B.S., Cao B. , Emerson J.J. , Chen C.  (2022). The 3D architecture of the pepper genome and its relationship to function and evolution traits. **Nature Communications** (2022) 13 3479.
  31. Durkin S.M., Chakraborty M., Abrieux A., Lewald K.M., Gadau A., Svetec N., Peng J., Kopyto M., Chiu J.C., **Emerson J.J.** , Zhao L.  Behavioral and genomic sensory adaptations underlying the pest activity of *Drosophila suzukii*. **Mol. Biol. Evol.** (2021) msab048.
  30. Chakraborty M.\*, Ramaiah A.\*, Adolphi A., Halas P., Kaduskar B., Ngo L.T., Jayaprasad S., Paul K., Whadgar S., Srinivasan S., Subramani S., Bier E., James A.A., **Emerson J.J.** . Hidden genomic features of an invasive malaria vector, *Anopheles stephensi*, revealed by a chromosome-level genome assembly. **BMC Biol.** (2021) 19 28.
  29. Liao Y. , Zhang X., Chakraborty M., **Emerson J.J.**  Topologically associating domains and their role in the evolution of genome structure and function in *Drosophila*. **Genome Research** (2021) 31: 397.
  28. Chakraborty M.\*, Chang C.-H.\*, Khost D.E., Vedanayagam J., Adrion J.R., Liao Y., Montooth K., Meiklejohn C.D., Larracuent A.M. , **Emerson J.J.**  (2021). Evolution of genome structure in the *Drosophila simulans* species complex. **Genome Research** 31: 380.
  27. Adolphi A., Gantz V.M., Jasinskiene N., Lee H.-F., Hwang K., Bulger E.A., Ramaiah A., Bennett J.B., Terradas G., **Emerson J.J.** , Marshall J.M., Bier E., James A.A.  Efficient population modification gene-drive rescue system in the malaria mosquito *Anopheles stephensi*. **Nature Communications** (2020) 11 5553.
  26. Kou Y.\*, Liao Y.\*, Toivainen T., Lv Y., Tian X., **Emerson J.J.** , Gaut B.S. , Zhou Y.  Evolutionary genomics of structural variation in Asian rice (*Oryza sativa*) domestication. **Mol. Biol. Evol.** (2020) msaa185.
  25. Heras J. , Chakraborty M., **Emerson J.J.** , German D.P. Genomic and biochemical evidence of dietary adaptation in a marine herbivorous fish. **Proc. R. Soc. B.** (2020) 287.
  24. Chakraborty M., **Emerson J.J.** , Macdonald S.J., Long A.D. Structural variants exhibit allelic heterogeneity and shape variation in complex traits. **Nature Communications** (2019) 4872.
  23. Zhang X. and **Emerson J.J.**  Inferring Compensatory Evolution of *cis*- and *trans*-Regulatory Variation. **Trends in Genetics** (2019) 35 1.
  22. Solares E.A.\*, Chakraborty M.\*, Miller D.E., Kalsow S., Hall K.E., Perera A.G., **Emerson J.J.** , Hawley R.S.  Rapid low-cost assembly of the *Drosophila melanogaster* reference genome using low-coverage, long-read sequencing. **G3: Genes, Genomes, Genetics** (2018) 8 3143.
  21. Chakraborty M. , VanKuren N.W. Zhao R., Zhang X., Kalsow S., and **Emerson J.J.** , Hidden genetic variation shapes the structure of functional elements in *Drosophila*. **Nature Genetics** 2018; 50, 20.
  20. **Emerson J.J.** , Evolution: A Paradigm Shift in Snake Sex Chromosome Genetics. **Current Biology** (2017); 27, R800.
  19. Long M. , **Emerson J.J.** , Meiotic Sex Chromosome Inactivation: Compensation by Gene Traffic. **Current Biology** (2017) 27, R659.
  18. Chakraborty M.\*, Baldwin-Brown J.G.\*, Long A.D., **Emerson J.J.**  Contiguous and accurate *de novo* assembly of metazoan genomes with modest long read coverage. **Nucleic Acids Research** (2016) 44, e147.
  17. Viçoso B.\*, **Emerson J.J.** , 2 others, Doris Bachtrog. Comparative Sex Chromosome Genomics in Snakes: Differentiation, Evolutionary Strata, and Lack of Global Dosage Compensation. **PLoS Biology** (2013) 11:e1001643.
  16. Schaefer B.\*, **Emerson J.J.** , 3 others, Li W.H. Inheritance of gene expression level and selective constraints on trans- and cis-regulatory changes in yeast. **Mol. Biol. Evol.** (2013) 30(9):212.
  15. Pool J.E., Corbett-Detig R.B., Sugino R.P., Stevens K.A., Cardeno C.M., Crepeau M.W., Duchon P., **Emerson J.J.** , Saelao P., Begun D.J., Langley C.H. Population Genomics of sub-saharan *Drosophila melanogaster*: African diversity and non- African admixture. **PLoS Genet.** (2012) 8(12):e1003080.
  14. Cardoso-Moreira M., **Emerson J.J.** , *et al.* *Drosophila* duplication hotspots are associated with late- replicating regions of the genome. **PLoS Genet.** (2011) 7(11): e1002340.

13. Nikaido M., Sasaki T., **Emerson J.J.**, 6 others, Li W.-H., Okada N. Genetically distinct coelacanth population off the northern Tanzanian coast. **Proc. Natl. Acad. Sci. U.S.A.** (2011) 108(44): 18009.
12. **Emerson J.J.\***, Hsieh L.- C.\*, Sung H.- M.\*, Wang T.- Y.\*, 4 others, Li W.- H. Natural selection on cis and trans regulation in yeasts. **Genome Research** (2010) 20(6): 826.
11. **Emerson J.J.**, Li W.- H. The genetic basis of evolutionary change in gene expression levels. **Phil. Trans. R. Soc. B.** (2010) 365(1552): 2581.
10. **Emerson J.J.\***, Cardoso- Moreira M.\*, Borevitz J.O., Long M. Natural selection shapes genome- wide patterns of copy- number polymorphism in *Drosophila melanogaster*. **Science.** (2008) 320(5883): 1629.
9. Fan Chuanzhu, **Emerson J.J.**, Manyuan Long, M. The origin of new genes. **Evolutionary Genomics and Proteomics**, Mark Pagel & Andrew Pomiankowski (eds.) (2007) Sinauer Associates, Inc., Sunderland, Massachusetts, USA. pp27.
8. Chen Y.\*, **Emerson J.J.\***, Martin T.M.\* Codon volatility does not detect selection. **Nature.** (2005) 433:E6-7.
7. Cowan A.T., Bowman G.R., Edwards K.F., **Emerson J.J.**, Turkewitz A.P. Genetics, genomic, and functional analysis of the granule lattice proteins in *Tetrahymena* secretory granules. **Mol. Biol. Cell.** (2005) 17(9): 4046.
6. International Chicken Genome Sequencing Consortium. Sequence and comparative analysis of the chicken genome provide unique perspectives on vertebrate evolution. **Nature.** (2004) 432:695.
5. **Emerson J.J.\***, Kaessmann H.\*, Betrán E., Long M. Extensive gene traffic on the mammalian X chromosome. **Science.** (2004) 303(5657): 537.
4. Betrán E., **Emerson J.J.**, Kaessmann H., Long M. Sex Chromosomes and Male Functions: Where Do New Genes Go? **Cell Cycle.** (2004) 3(7):873.
3. Wang W., Thornton K., **Emerson J.J.**, Long M. Nucleotide variation and recombination along the fourth chromosome in *Drosophila simulans*. **Genetics.** (2004) 166(4): 1783.
2. Strassmann J.E., Queller D.C., **Emerson J.J.**, Stagi M., Cervo R., Turillazzi S. Comparing the costs and benefits of grouping with non- relatives in the social amoeba *Dictyostelium discoideum* (Amoebozoa) and the social wasp *Polistes dominulus* (Hymenoptera vespidae). **Redia**, LXXXVII, (2004) 145.
1. Bergelson J., Dwyer G., **Emerson J.J.** Models and data on plant- enemy coevolution. **Annu. Rev. Genet.** (2001) 35:469.

## EXTERNAL FUNDING

- 2025-2029 Co-PI NSF \$1,691,891 Co-PI with Brandon Gaut  
Plant Genome Research Program: Genome evolution in 2D and 3D
- 2024-2029 PI NIH \$1,896,915 Sole PI  
The origin, maintenance, and adaptive consequences of variation in genome structure
- 2017-2023 PI NIH \$1,506,952 Sole PI  
Structural variation, population size, and the evolution of genome complexity
- 2017-2023 Co-PI NSF \$855,590 Co-PI with Adriana Briscoe  
Mechanisms of Color Vision: Genomics, Physiology and Behavior

## CONFERENCE AND SOCIETY TALKS (SINCE 2010)

1. 07/20/2020, Arthropod Genomics Conference (virtual), Evolution of genome structure and function in *Drosophila*.
2. 11/05/2019, TIGS Active Genetics Annual Symposium, Genome structure and variation in mosquitoes and flies: Hidden, heterogeneous, and functional
3. 8/16/2021, AsiEvo Conference, A W-linked gene duplication underlies sexually dimorphic UV color vision in *Heliconius* butterflies
4. 7/14/2014, National Centers for Systems Biology, Sanford Consortium for Regenerative Medicine, San Diego, California: Evolution of Complex Biological Systems.
5. 7/2013, Society for Molecular Biology and Evolution: Using Experimental Evolution Data to Improve Models of Phenotypic Adaptation.

## INVITED TALKS (SINCE 2010)

1. 5/27/2024, Center for Ecological and Evolutionary Dynamics Seminar, USC: "The ghost of Hubby &

Lewontin: Genetic variation in the uncharted wilds of the genome”.

2. 4/22/2022, Molecular and Computational Biology Seminar, USC: “The end of the beginning: Genome structural variation in the era of complete genomes”.
3. 4/2/2021, Bioinformatics Seminar UNC Charlotte, Genome structure and variation: Hidden, heterogeneous, and functional.
4. 3/9/2021, Vetmeduni Vienna Popgen Seminar Series, Genome structure and variation: Hidden, heterogeneous, and functional.
5. 7/23/2019, Keynote EMBO Workshop, Molecular and population biology of mosquitoes and other disease vectors: Genome structure and variation in mosquitoes and flies: Hidden, heterogeneous, and functional.
6. 6/12/2019, University of Houston: The evolution of genome structure in flies and fish.
7. 1/28/2019, University of California, San Diego: A new reference quality genome of the Asian malaria vector *Anopheles stephensi*.
8. 6/29/2018, Graduate University for Advanced Studies (Hayama, Kanagawa, Japan): The evolution of genome structure in *Drosophila*.
9. 4/25/2018, Loma Linda University: The evolution of genome structure in *Drosophila*.
10. 3/19/2018, Arizona State University: The Evolution of Genome Structure in *Drosophila*.
11. 1/24/2018, University of Nebraska, Lincoln: The Evolution of Genome Structure in *Drosophila*.
12. 6/19/2017, University of California, San Diego: Genomes to phenotype: Causes and consequences of structural variation in *Drosophila*.
13. 12/29/2016, National Taiwan University Medical School: Beneath the tip of the iceberg: using high quality genomes to uncover the evolutionary consequences of hidden genetic variation in *Drosophila*.
14. 12/28/2016, National Taiwan University Dept. Life Sciences: Beneath the tip of the iceberg: using high quality genomes to uncover the evolutionary consequences of hidden genetic variation in *Drosophila*.
15. 12/20/2016, Biodiversity Research Center, Academia Sinica, Taipei, Taiwan: Beneath the tip of the iceberg: using high quality genomes to uncover the evolutionary consequences of hidden genetic variation in *Drosophila*.
16. 9/13/2016, Cornell University: Beneath the tip of the iceberg: using high quality genomes to uncover the evolutionary consequences of hidden genetic variation in *Drosophila*.
17. 9/12/2016, University of Rochester: Beneath the tip of the iceberg: using high quality genomes to uncover the evolutionary consequences of hidden genetic variation in *Drosophila*.
18. 3/19/2015, Michigan State University: Evolution and novelty: exploring adaptation from the perspectives of experimental evolution and population genomics.
19. 6/19/2013, Pomona College (Claremont Colleges) HHMI Summer Colloquium: Unraveling sex chromosome differentiation in snakes using high throughput sequencing.
20. 3/2012, University of California, Irvine: Duplication, sex chromosomes, and regulation: population genetic perspectives on genome evolution.
21. 2/2012, Johns Hopkins University, Department of Biology: Duplication, sex chromosomes, and regulation: population genetic perspectives on genome evolution.
22. 2/1/2012, University of Texas, Southwestern Medical School: Duplication, sex chromosomes, and regulation: population genetic perspectives on genome evolution.
23. 6/1/2011, University of Wisconsin, Madison: Duplication, sex chromosomes, and regulation: population genetic perspectives on genome evolution.

## CONFERENCE ORGANIZATION

1. 2024, (Session Co-Chair). The Allied Genetics Conference Meeting 2024 (A Genetics Society of America Conference), Genomes and Genomics, National Harbor, Maryland.
2. 2019-present, (Founder and Co-Organizer). Southern California Evolutionary (SCaE) Genetics Meeting. Meeting at UCI announced, planned for Oct. 19, 2019 and approximately every 6 months thereafter.
3. 7/2015, SMBE Meeting 201 (Symposium Co-Chair), Genomics of sex bias: Addressing questions with or without genomes, Society for Molecular Biology and Evolution, Vienna, Austria.
4. 6/2014, (Symposium Co-Chair). SMBE Meeting 2014, Creative uses of next generation sequencing technology in evolutionary genomics: solving old problems with new approaches, Society for Molecular Biology and Evolution, San Juan, Puerto Rico.
5. 7/2013, (Symposium Co-Chair). SMBE Meeting 2013, Phenotypic Models of Adaptation and Experimental Data, Society for Molecular Biology and Evolution, Chicago, Illinois.

6. 2013-2015, (Founder and Co-Organizer). Southern California Evolutionary (SCalE) Genetics Meeting. Held 4 times (UCI, UCR, USC, UCLA).

## **SOCIETY MEMBERSHIPS**

- Society for Molecular Biology and Evolution
- Genetics Society of America

## **JOURNAL PEER REVIEW SERVICE (2016-PRESENT)**

BMC Biology, Current Biology, Developmental Biology, eLife, Evolution, G3, Genetics, Genome Biology and Evolution, Genome Research, Molecular Biology and Evolution, Nature Methods, Peer Community in Evolutionary Biology, PLoS Biology, PNAS, Phil. Trans. Roy. Soc. B

## **JOURNAL EDITORIAL REVIEW SERVICE**

Associate editor for the journal Genome Biology and Evolution  
Associate editor for the journal G3: Genes | Genomes | Genetics

## **GRANT PROPOSAL REVIEW SERVICE**

- NIH R01 NHGRI Panel (2023, 2025)
- NSF CAREER Award (2017, 2018)
- NIH U-type/Cooperative Agreements Panel (2015)